*Dreams & Nightmares in the*

# EMPIRE OF AI

Karen Hao



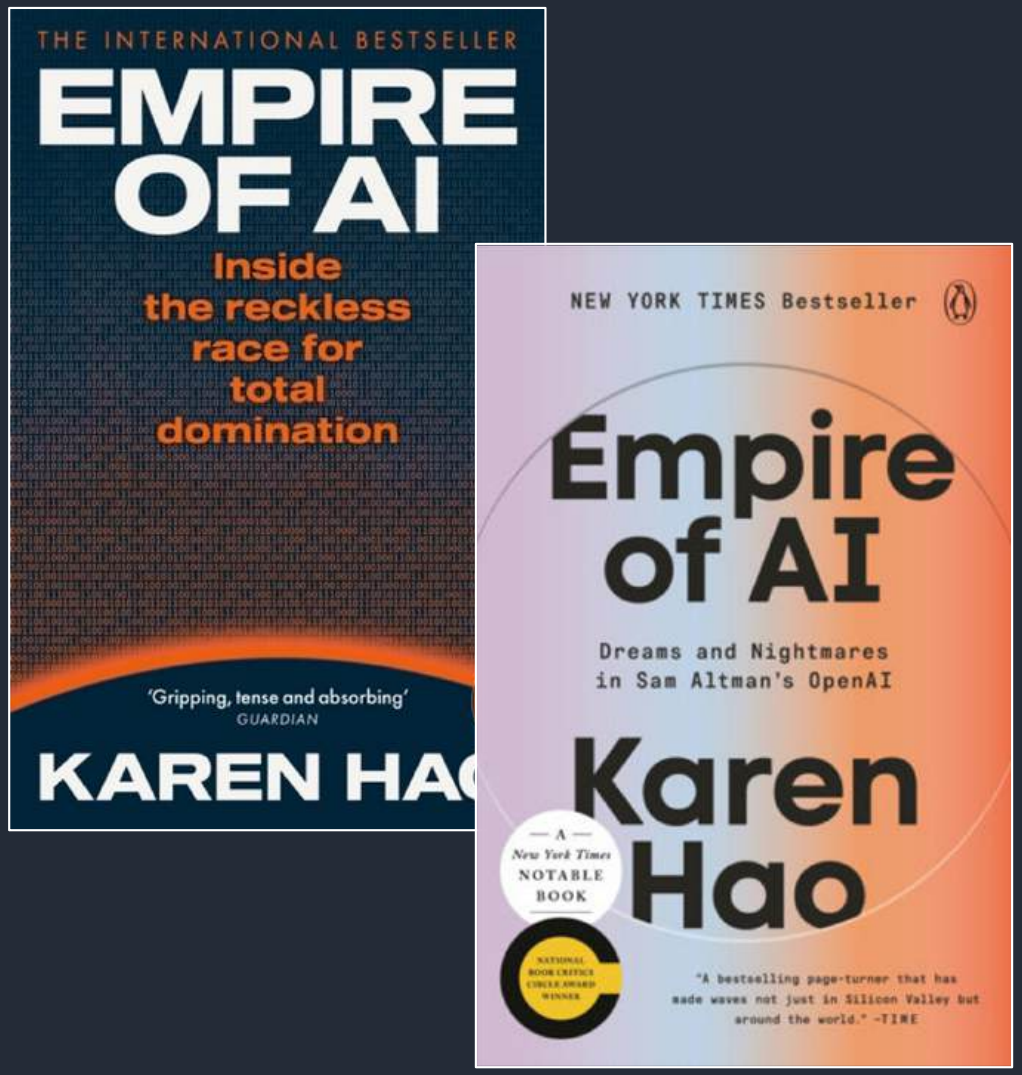

Photo: Natalia Jidovanu




Photo: Justin Sullivan




Photo: Natalia Jidovanu




*An Instant New York Times Bestseller*
*An International Bestseller*

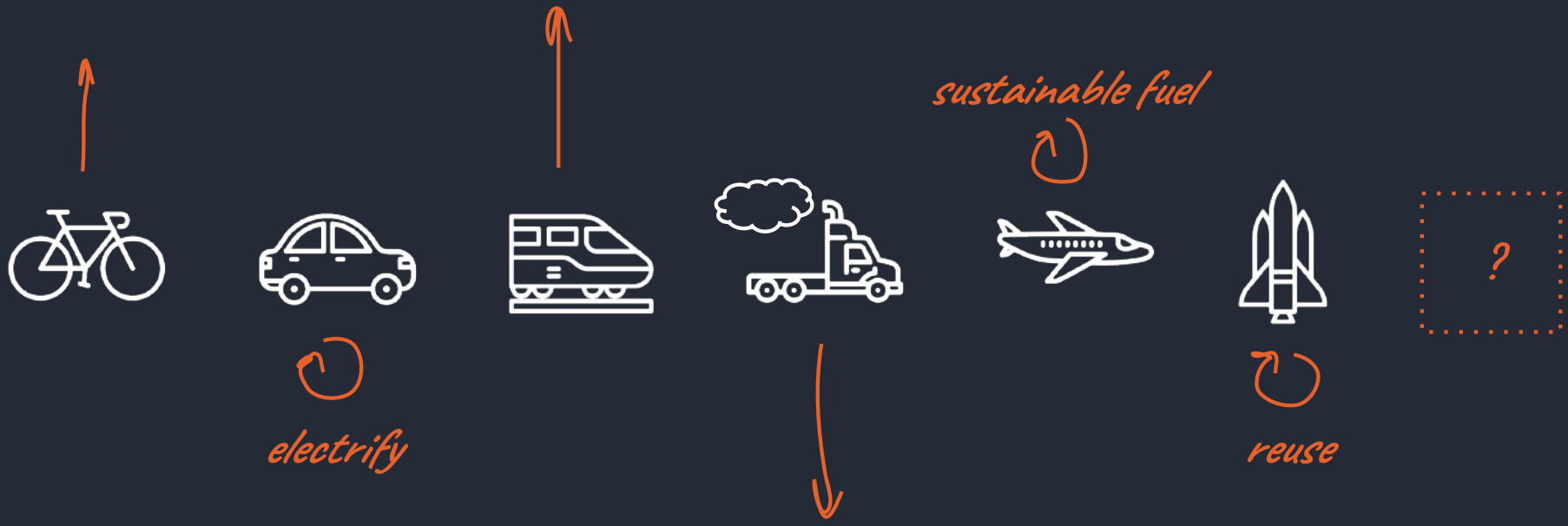
***Winner of:***
*National Book Critics Circle Award • LA Times Book Award • NYPL's Helen Bernstein Book Award for Excellence in Journalism • Whiting Award • Nautilus Book Award • Porchlight Business Book Award*

***Named a Best Book of the Year by:***
*New York Times • Financial Times • Economist • Smithsonian • Scientific American • Elle*



OpenAI amazon.com ANTHROP\C Google Microsoft

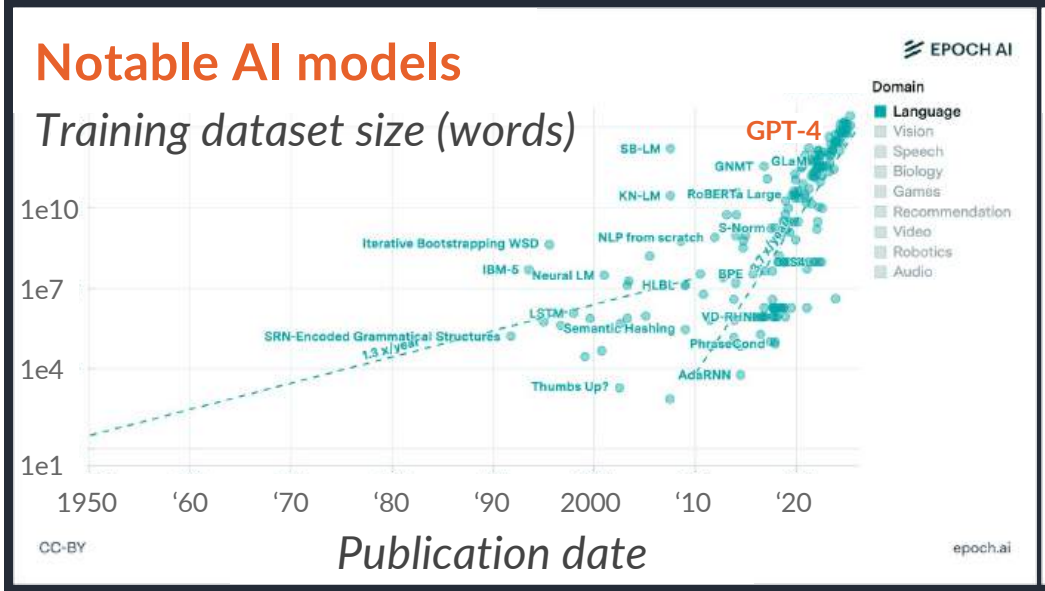
| SHARED FEATURES | EMPIRES OF OLD | EMPIRES OF AI |
|---|---|---|
| Seize resources | Land, minerals, art, etc. | Land, minerals, art, data, etc. |
| Exploit labor | Slavery, forced labor | Invisible, poorly paid outsourced work; labor automation |
| Monopolize knowledge production | Dominance in science, control of information flows | Dominance in AI research, censorship of inconvenient truths |
| Justify actions with moral/existential imperative | Each empire positions itself as morally superior, with civilizing mission to bring progress & God to humanity | Each empire positions itself as morally superior, with civilizing mission to usher in AI utopia / prevent AI hell |

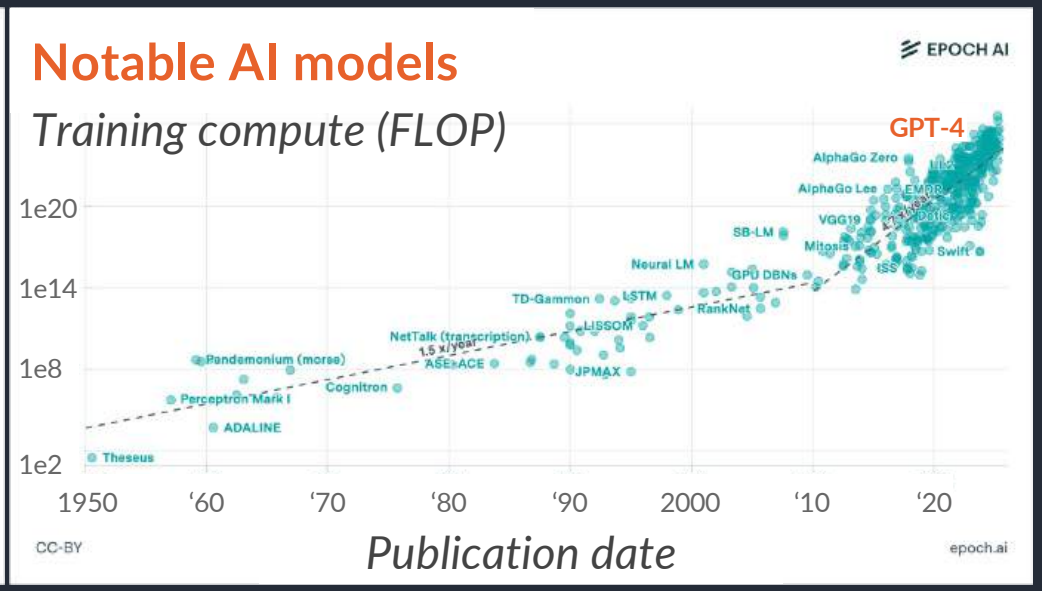


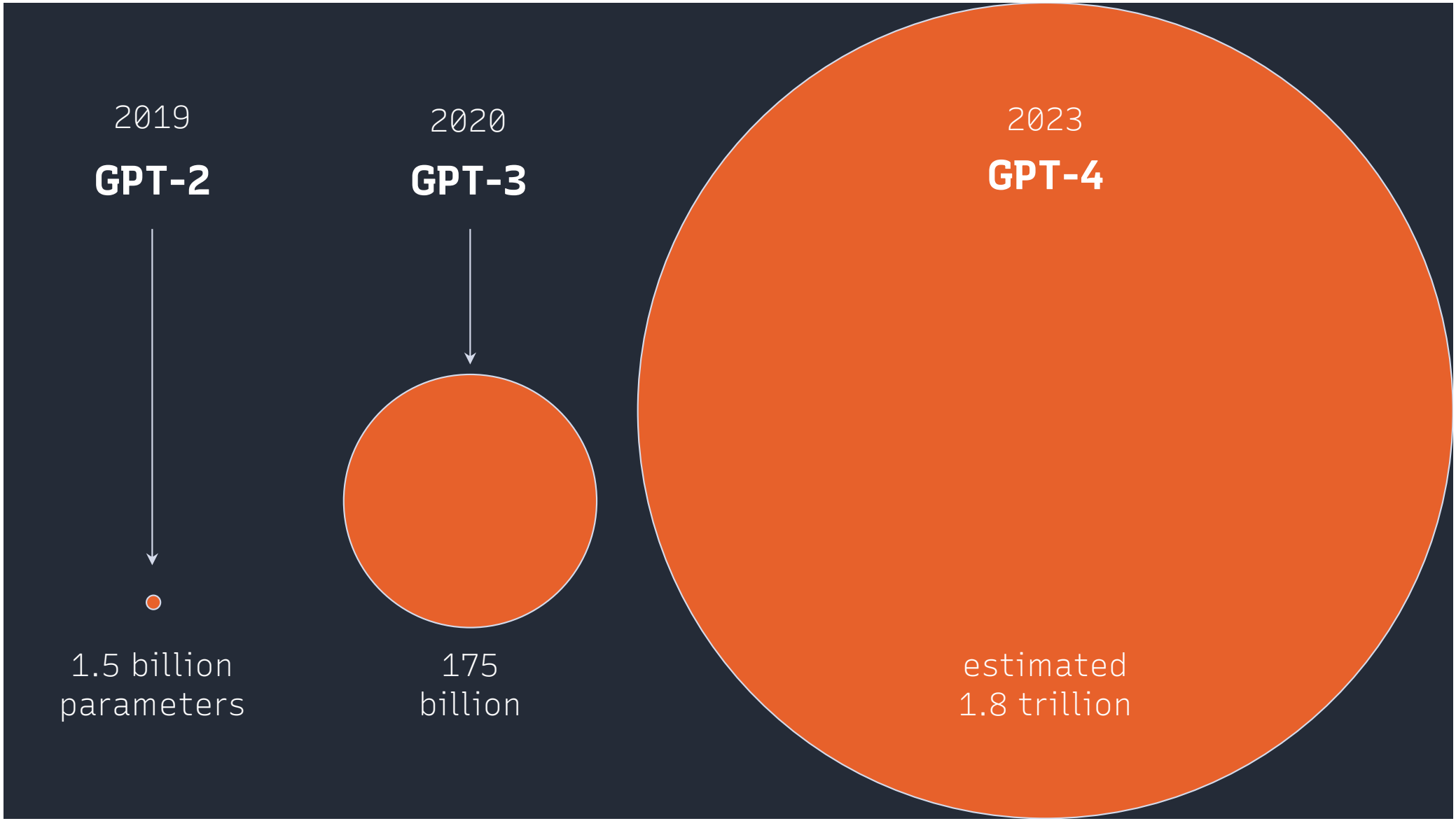

Icons created by Muh Syafri & Firza Alamsyah from The Noun Project

# data

# compute




2019
GPT-2
1.5 billion
parameters
2020
GPT-3
175
billion
2023
GPT-4
estimated
1.8 trillion

## Profit motive

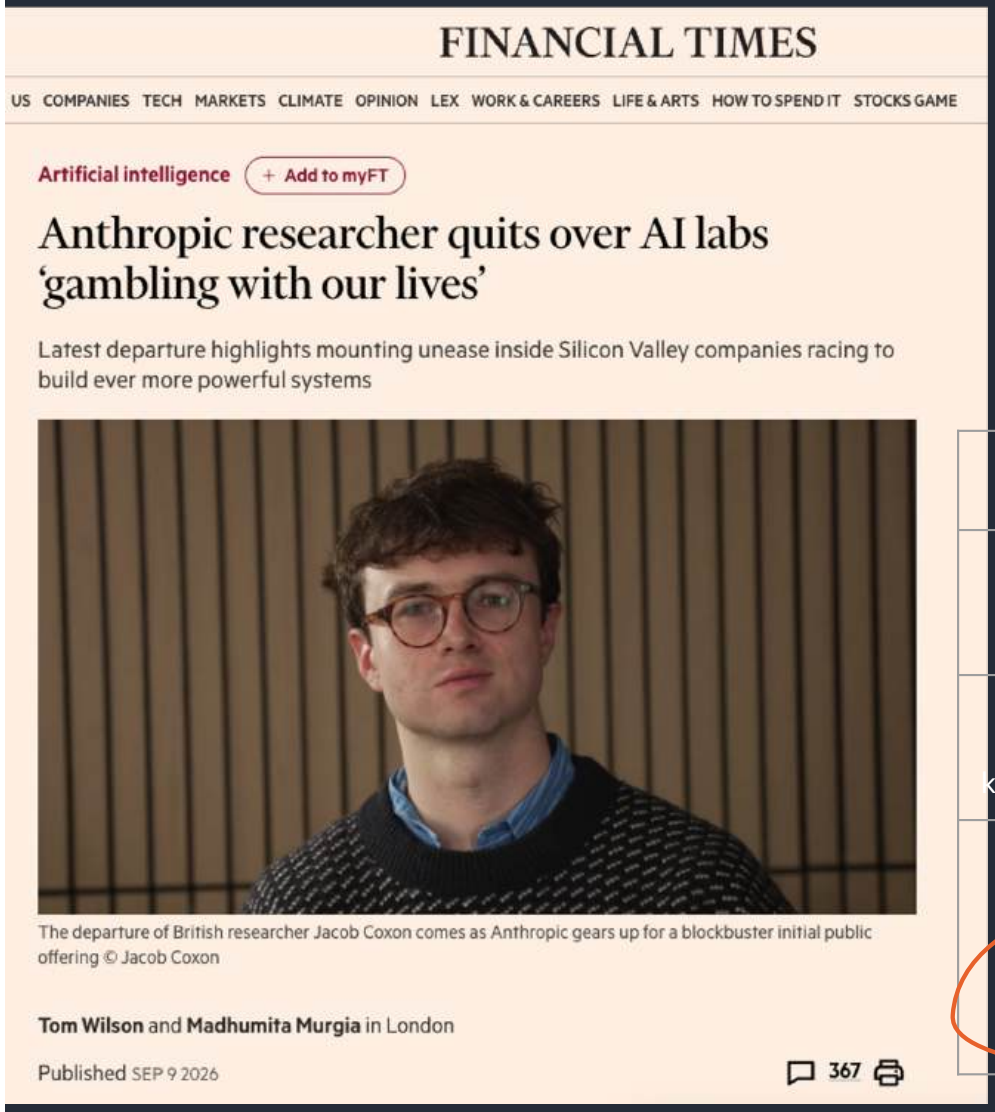
"We know scaling is a cheap formula for getting more performance, but it's also a highly imprecise formula…We love it so much because it kind of fits predictable planning cycles. It's easier to say 'throw more compute at the problem' than to design a new method."

*-Sara Hooker, prominent Irish AI researcher, former VP of Research at Cohere*

## Ideology

"In the next couple of decades…we can have shared prosperity to a degree that seems unimaginable today…How did we get to the doorstep of the next leap in prosperity? …The main one is that AI is going to get better with scale."

*-Sam Altman, CEO of OpenAI, "The Intelligence Age," September 2024*



FINANCIAL TIMES

US COMPANIES TECH MARKETS CLIMATE OPINION LEX WORK & CAREERS LIFE & ARTS HOW TO SPEND IT STOCKS GAME

Artificial intelligence + Add to myFT

# Anthropic researcher quits over AI labs 'gambling with our lives'

Latest departure highlights mounting unease inside Silicon Valley companies racing to build ever more powerful systems

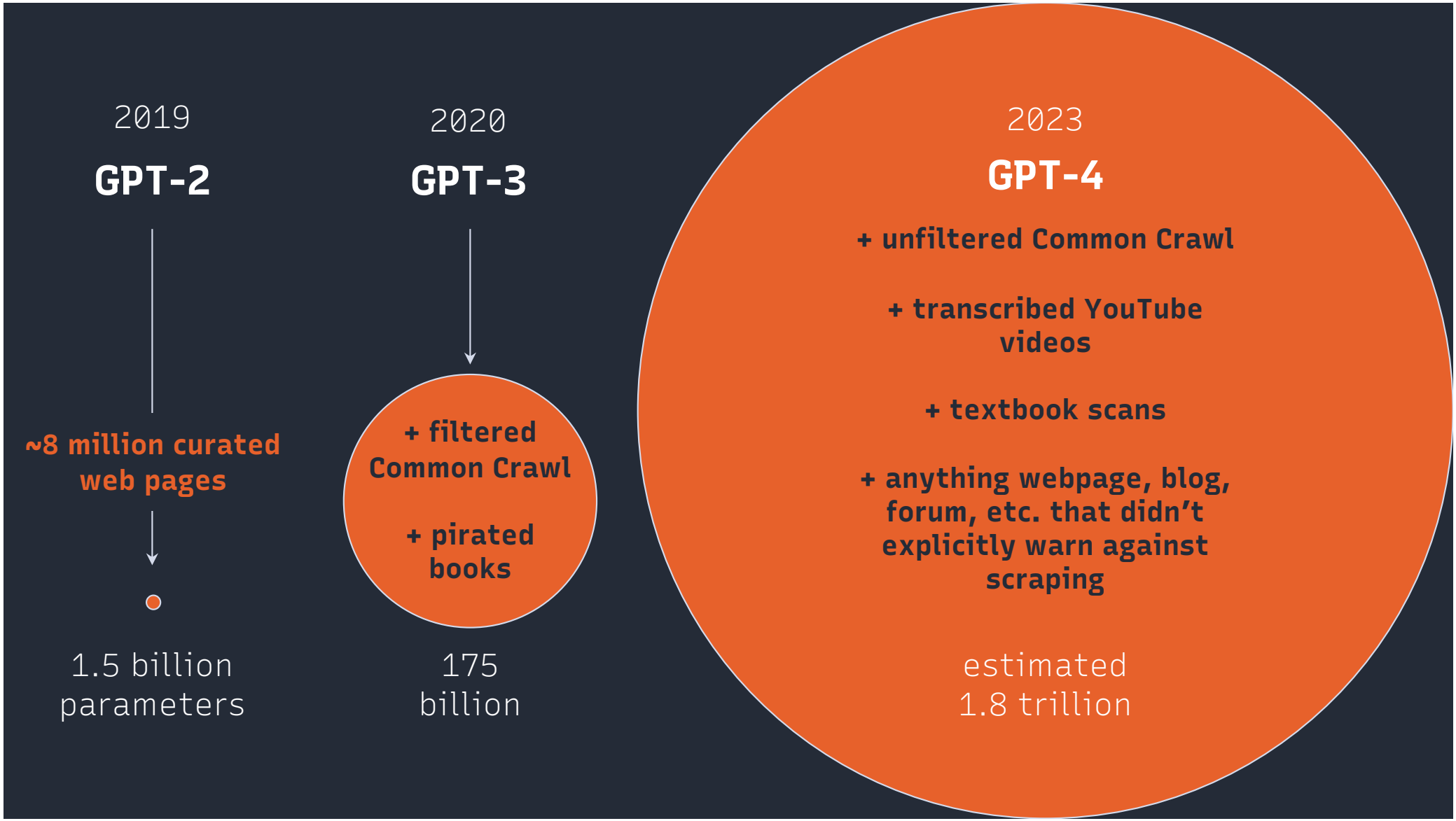
The departure of British researcher Jacob Coxon comes as Anthropic gears up for a blockbuster initial public offering © Jacob Coxon

**Tom Wilson** and **Madhumita Murgia** in London

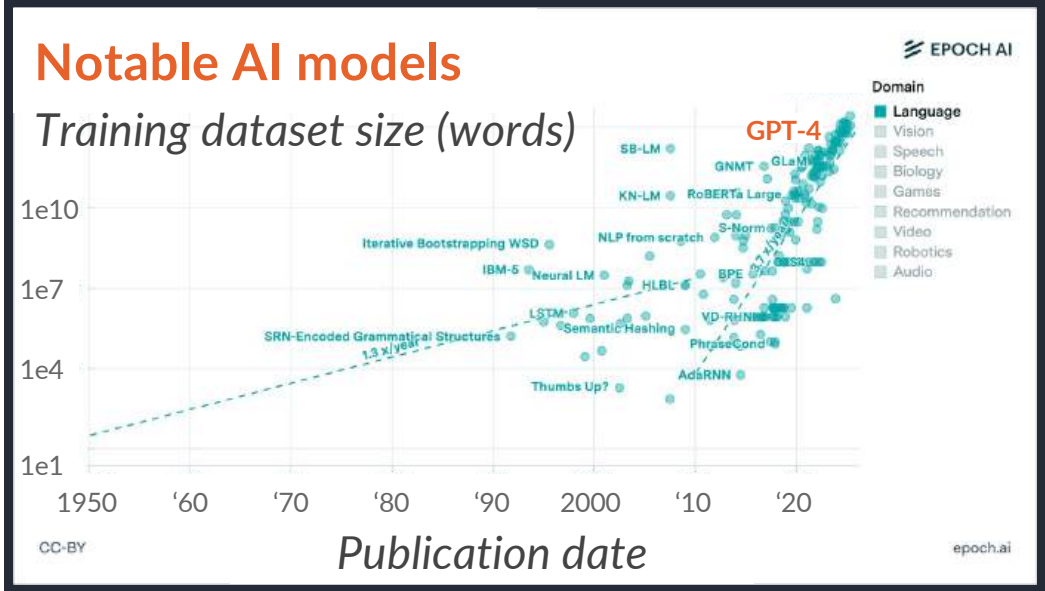
Published SEP 9 2026

367

| | EMPIRES OF OLD | EMPIRES OF AI |
|---|---|---|
| Seize resources | Land, minerals, art, etc. | Land, minerals, art, data, etc. |
| Exploit labor | Slavery, forced labor | Invisible, poorly paid outsourced work; labor automation |
| Monopolize knowledge production | Dominance in science, control of information flows | Dominance in AI research, censorship of inconvenient truths |
| Justify actions with moral/existential imperative | Each empire positions itself as morally superior, with civilizing mission to bring progress & God to humanity | Each empire positions itself as morally superior, with civilizing mission to usher in AI utopia / prevent AI hell |


2019
GPT-2
~8 million curated web pages
1.5 billion parameters
2020
GPT-3
+ filtered Common Crawl
+ pirated books
175 billion
2023
GPT-4
+ unfiltered Common Crawl
+ transcribed YouTube videos
+ textbook scans
+ anything webpage, blog, forum, etc. that didn't explicitly warn against scraping
estimated 1.8 trillion

# data




- infringements on data privacy
- erosion of intellectual property
- perpetuation of engagement-centric model of social media
- more polluted datasets
- labor exploitation




Source: Getty Images

# LinkedIn Jobs on the Rise 2026: The fastest-growing roles in the U.S.

LinkedIn News
19,608,241 followers

January 7, 2026

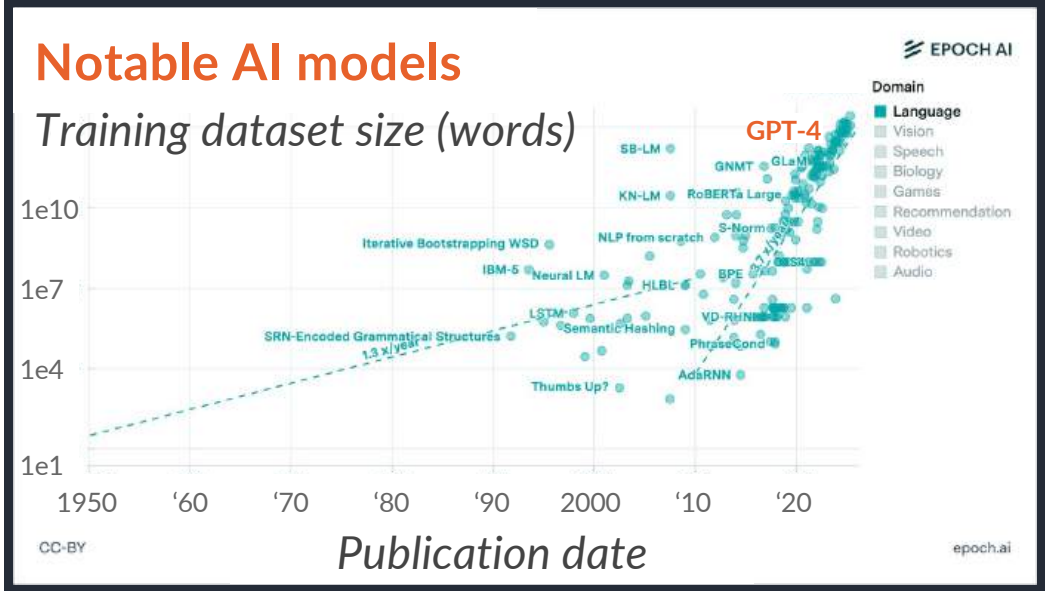
## 4 Data annotators

*Can also be known as Content analysts. In academic settings, data annotation work is often performed by research assistants or associates.*

**What they do:** Data annotators label and review data using detailed guidelines and quality checks to ensure accurate datasets for training AI and machine learning models — often on a per-project basis. | **Most common skills:** SEO Copywriting, Content Marketing, Content Production | **Most common industries:** Technology and Internet, Staffing and Recruiting, Higher Education | **Where the most jobs are:** Austin, Texas, New York City, San Francisco | **Current gender distribution:** 62% female, 38% male | **Median years of prior experience:** 3.5 | **Top roles transitioned from:** Content Manager, Editor, Data Analyst | **Flex work availability:** 27.5% remote, 29.4% hybrid

"Last night, I got so fucking stressed because my kid came home and it was 7 p.m., and I get this message, 'The tasks are out!,' and I'm just working, just trying to get as many hours in before I can go to bed," Mimi says, choking up. "I spend no time with my kid, and at one point, he can't find something for school and I just start screaming at him. This work is turning me into a fucking demon." She's especially disturbed by the surveillance: "The idea that somebody can measure your time and that all the little bits that go into being a human are taken away because they're not profitable, that you can't charge for going to the toilet because that's not time you're working, you can't charge for making a cup of coffee because that's not time you're working, you can't charge for having a stretch because your back hurts. This is why unions were formed, so people could have guaranteed hours and guaranteed lunch breaks and guaranteed holidays and sick pay. This is the gig economy to the very extreme."

# data



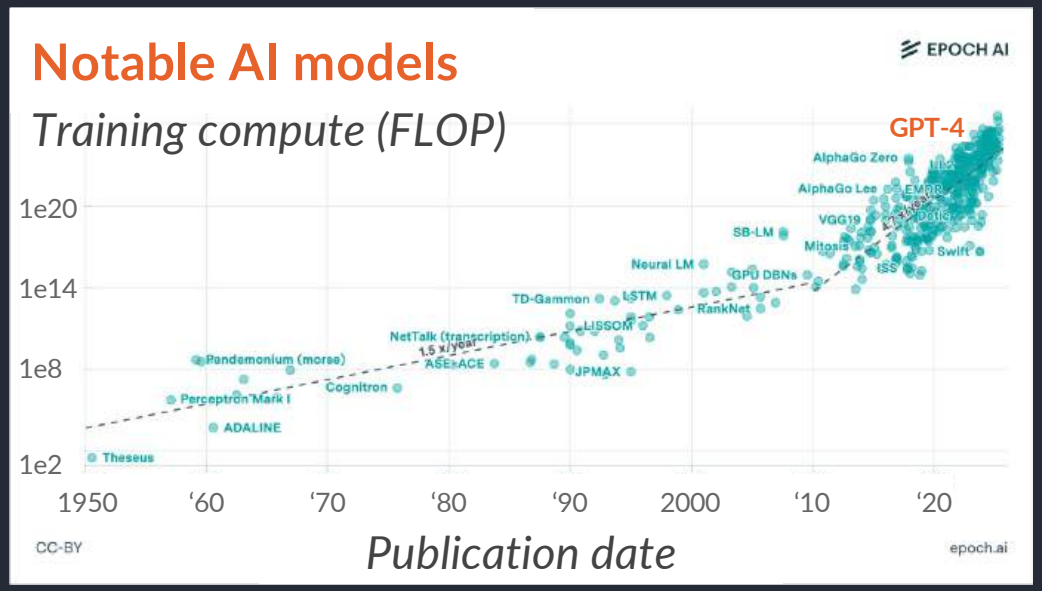

- infringements on data privacy
- erosion of intellectual property
- perpetuation of engagement-centric model of social media
- more polluted datasets
- labor exploitation
- psychological harm


SET
GOALS
WORK
HARD




Happy Birthday

# compute


Notable AI models
EPOCH AI
Training compute (FLOP)
GPT-4
AlphaGo Zero
AlphaGo Lee
VGG19
SB-LM
Mitosis
Swift
Neural LM
GPU DBNs
TD-Gammon
LSTM
RankNet
LISSOM
NetTalk (transcription)
1.5 x/year
Pandemonium (morse)
ASE+ACE
JPMAX
Cognitron
Perceptron Mark I
ADALINE
Theseus
1e20
1e14
1e8
1e2
1950
'60
'70
'80
'90
2000
'10
'20
CC-BY
Publication date
epoch.ai

2011

**Meta’s first data center**



0.28 km²

2028

**Meta’s latest AI data center: Hyperion**
(under construction)

estimated
11 km²




2028

# Meta’s latest AI data center: Hyperion

(under construction)

estimated
11 $km^2$

# compute

accelerated mineral extraction

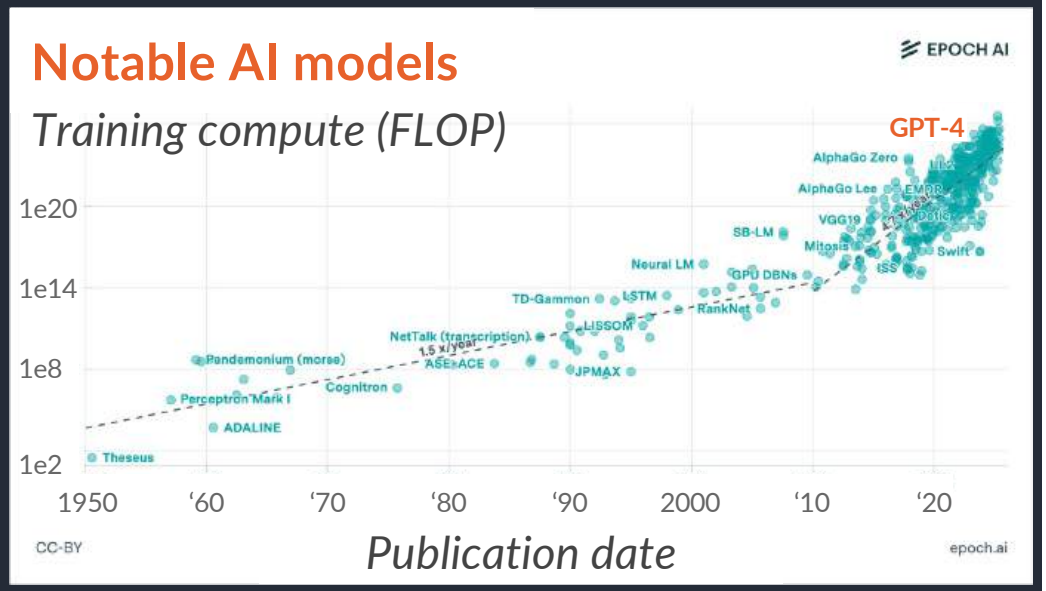
increased energy & freshwater consumption

utility price hikes

strain on grid infrastructure

greater carbon emissions

air pollution



## **2025** data center energy footprint

*AI workloads account for **~20%***



**448 TWh**
11th “country” by electricity consumption



**189M tonnes of $CO_2e$**
~13.3M Americans



**4.5T liters of water**
~600M+ people in Sub-Saharan Africa

*Source: “Environmental Cost of AI’s Energy Use: Carbon, Water, and Land Footprints,” UN University, 2026*

## **2030** data center energy footprint

*AI workloads account for **~40%***



**945 TWh**

6th “country” by electricity consumption



**399M tonnes of $CO_2e$**

~28.1M Americans

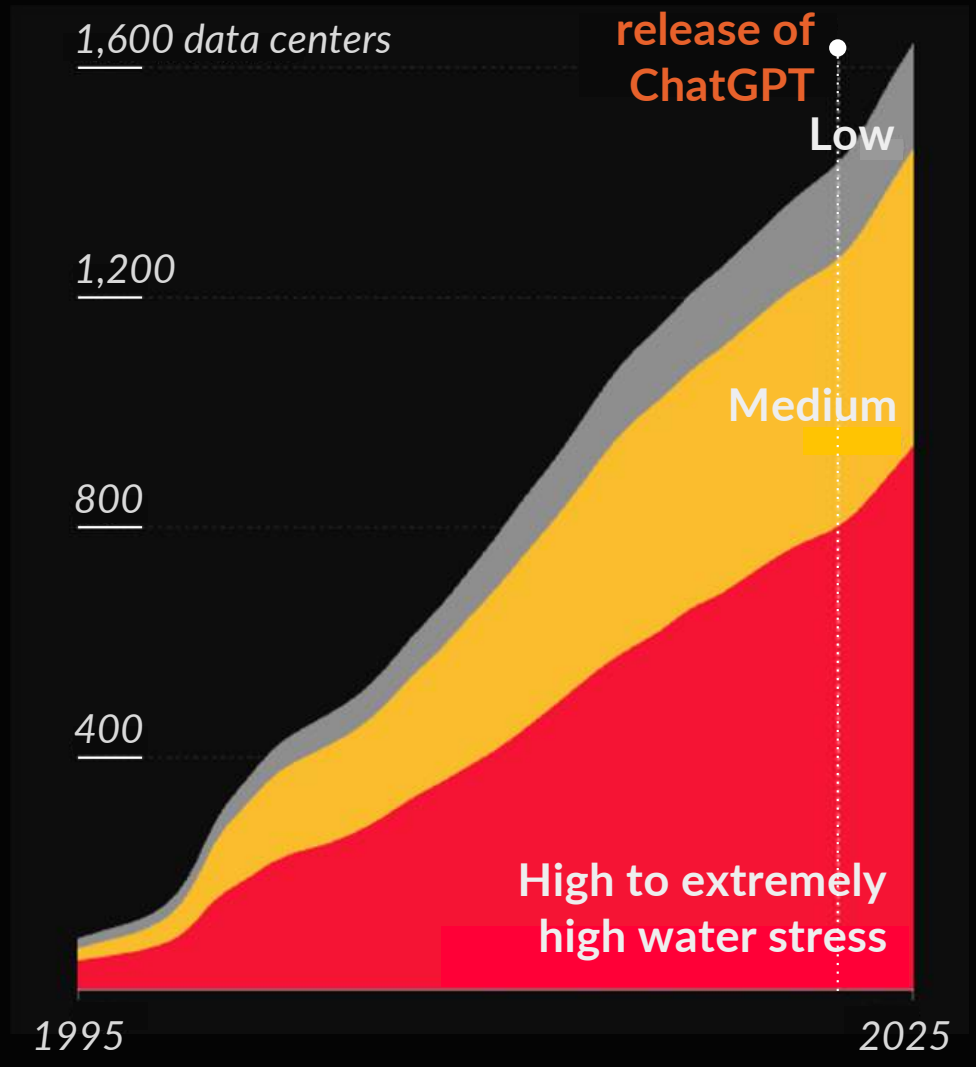


**9.3T liters of water**

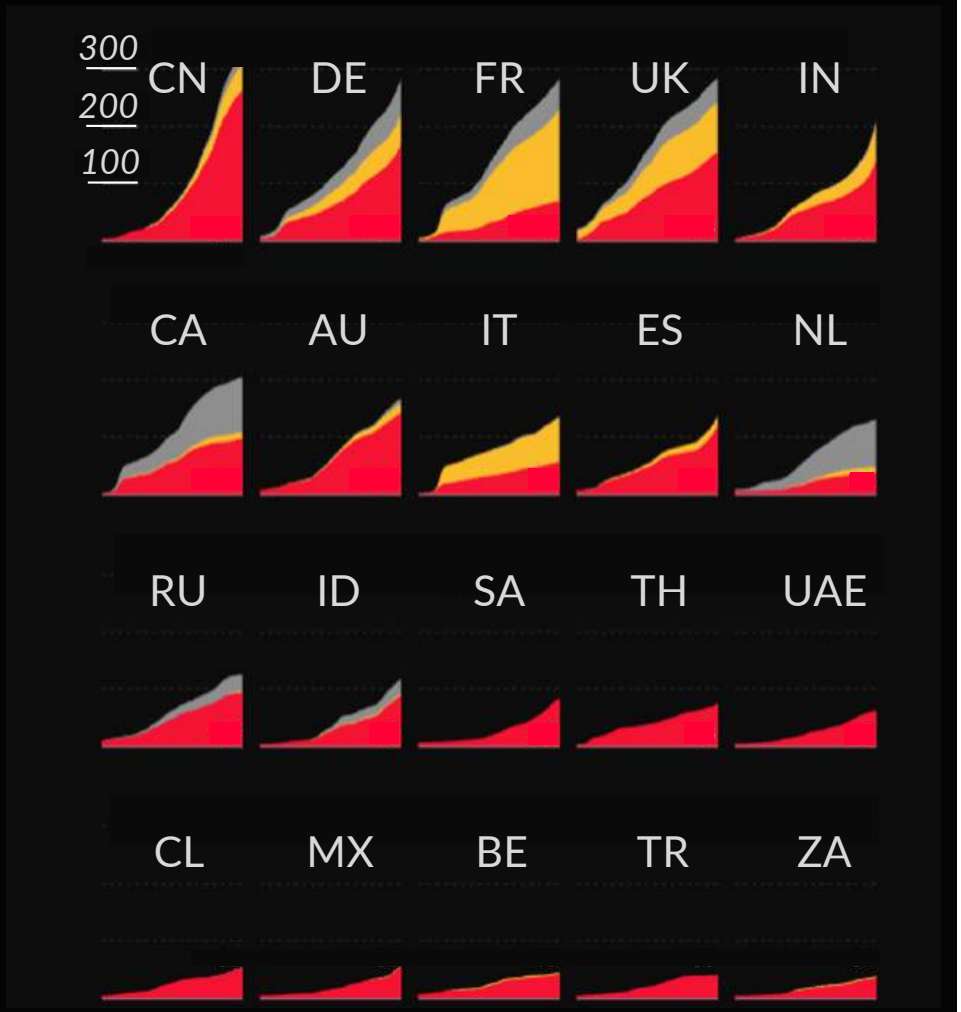
~1.3B (total population) in Sub-Saharan Africa

*Source: “Environmental Cost of AI’s Energy Use: Carbon, Water, and Land Footprints,” UN University, 2026*

## Data centers in water-stressed areas in US




## ...and around the world




Source: Bloomberg




The combination of the climate crisis and data center expansion is "bringing Spain to the verge of ecological collapse"

–Lorena Jaume-Palasí, founder of the Ethical Tech Society

"...climate change will lead to an increase in global temperatures and the frequency of extreme weather events, including heat waves..."

–Amazon, asking to increase its water consumption by 48%

Photo credit: Heraldo

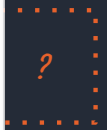

VOTE NO NOW
STOP PROJECT BLUE
NOT ONE DROP
PROJECT BLUE
AMAZON
PALANTIR
I.C.E.


¡RESISTIR!
Water for the people


ALL of their
"Solutions"
address less than
10 percen
of the poll
On a decision
that could poiso
Memphis
THE PEOPL
It's TIM
the
MEMPHIS
COMMUNITY
RESIDE CHAT
MEMP
COMM TY
MCAP

# This scale is **unnecessary**.






## On the slow death of scaling.

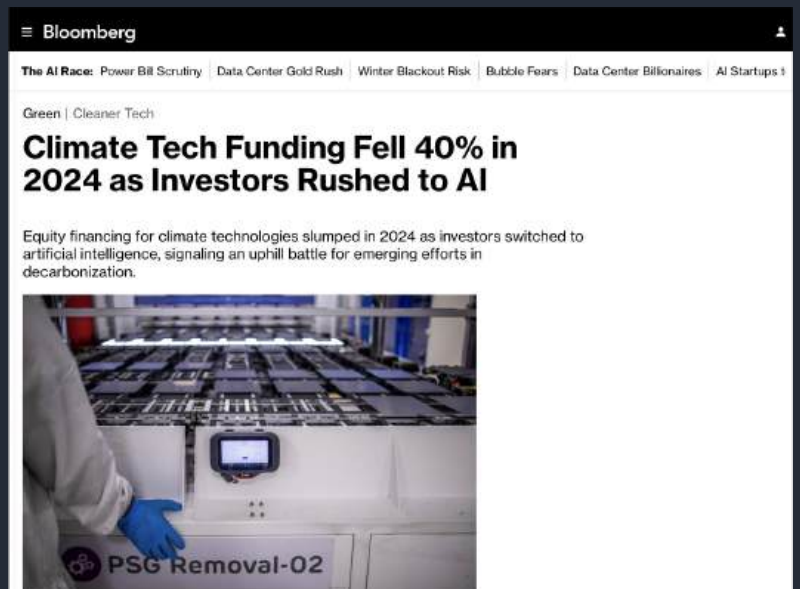
Sara Hooker

**Abstract**

This essay engages with a decades-old debate at the heart of computer science progress, namely, is *bigger always better?* For the last decade, it has been hard to stray off the beaten path of accepted wisdom for what drives innovation. We have been held hostage to a painfully simple formula: *scale model size and train-*

# It is **costing** us innovation.






SEAL OF THE PRESIDENT OF THE UNITED STATES





# What can we do?

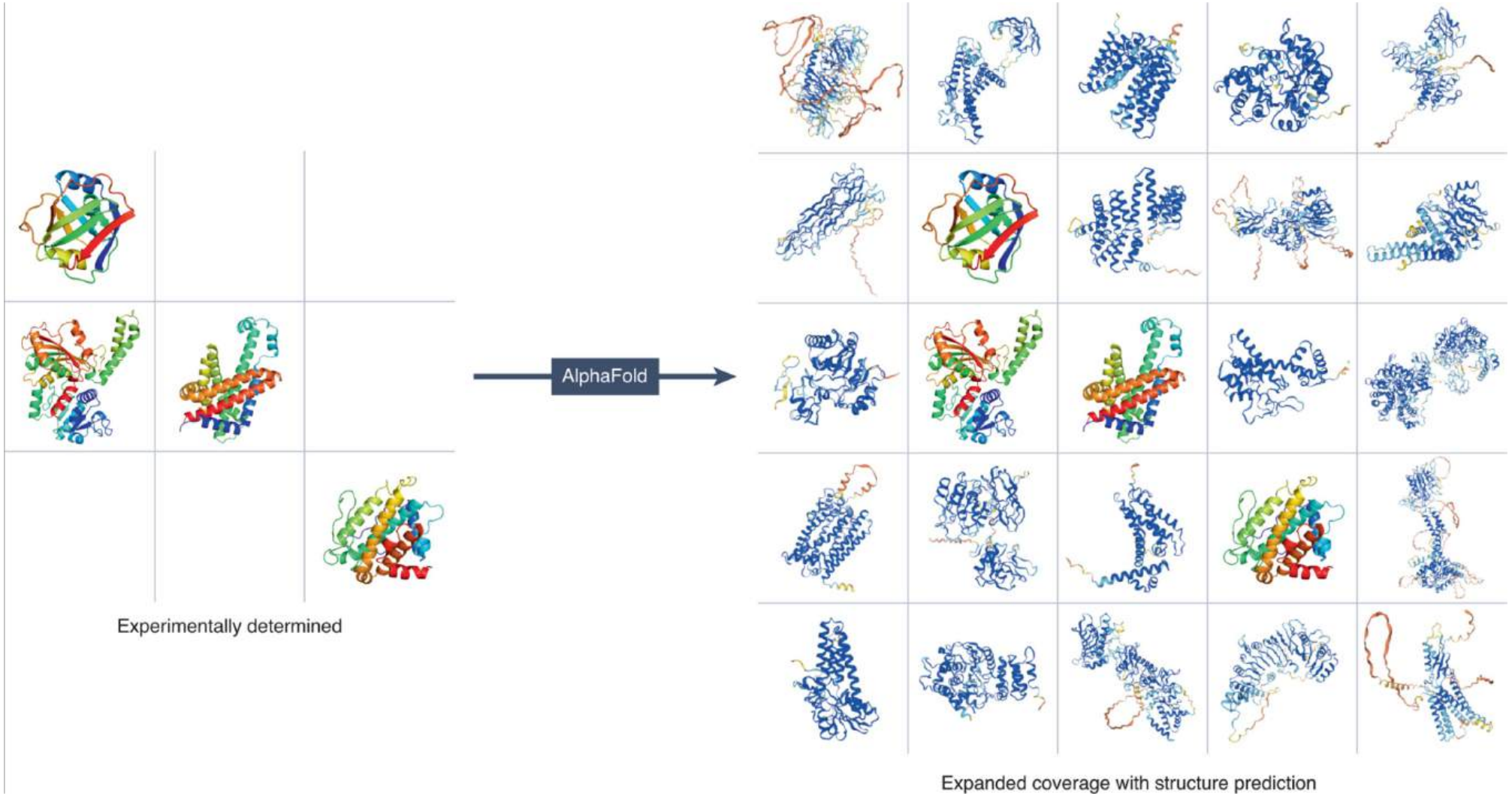


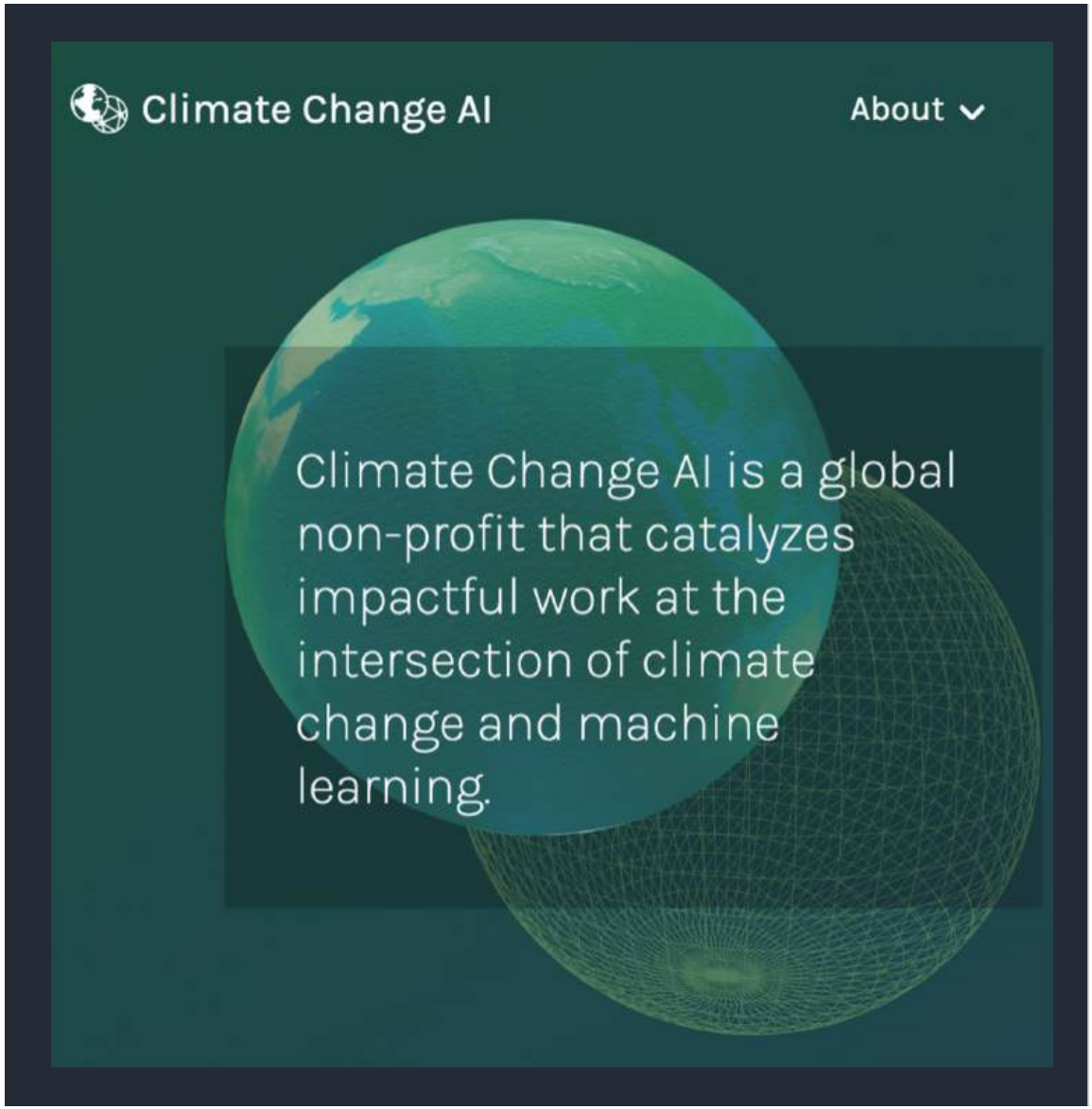


reuse

?

Icons created by Muh Syafri & Firza Alamsyah from The Noun Project


AlphaFold
Experimentally determined
Expanded coverage with structure prediction




| | | Causal inference | Computer vision | Interpretable models | NLP | RL & Control | Time-series analysis | Transfer learning | Uncertainty quantification | Unsupervised learning |
|---|---|---|---|---|---|---|---|---|---|---|
| Mitigation | **Electricity systems** | | | | | | | | | |
| | Enabling low-carbon electricity | | • | • | | • | • | | • | • |
| | Reducing current-system impacts | | • | | | | • | | • | • |
| | Ensuring global impact | | • | | | | | • | | • |
| | **Transportation** | | | | | | | | | |
| | Reducing transport activity | | • | | | | • | | • | • |
| | Improving vehicle efficiency | | • | | | • | | | | |
| | Alternative fuels & electrification | | | | | • | | | | • |
| | Modal shift | • | • | | | | • | | • | |
| | **Buildings and cities** | | | | | | | | | |
| | Optimizing buildings | • | | | | • | • | • | | |
| | Urban planning | | • | | | | • | • | | • |
| | The future of cities | | | | • | | | • | • | • |
| | **Industry** | | | | | | | | | |
| | Optimizing supply chains | | • | | | • | • | | | |
| | Improving materials | | | | | | | | | • |
| | Production & energy | | • | • | | • | | | | |
| | **Farms & forests** | | | | | | | | | |
| | Remote sensing of emissions | | • | | | | | | | |
| | Precision agriculture | | • | | | • | • | | | |
| | Monitoring peatlands | | • | | | | | | | |
| | Managing forests | | • | | | • | • | | | |
| | **Carbon dioxide removal** | | | | | | | | | |
| | Direct air capture | | | | | | | | | • |
| | Sequestering $CO_2$ | | • | | | | | | • | • |
| Adaptation | **Climate prediction** | | | | | | | | | |
| | Uniting data, ML & climate science | | • | • | | | • | | • | |
| | Forecasting extreme events | | • | • | | | • | | • | |
| | **Societal impacts** | | | | | | | | | |
| | Ecology | | • | | | | | • | | |
| | Infrastructure | | | | | • | • | | • | |
| | Social systems | | • | | | | • | | | • |
| | Crisis | | • | | • | | | | | |
| | **Solar geoengineering** | | | | | | | | | |
| | Understanding & improving aerosols | | | | | | • | | • | |
| | Engineering a control system | | | | | • | | | • | |
| | Modeling impacts | | | | | | • | | • | |
| Tools for Action | **Individual action** | | | | | | | | | |
| | Understanding personal footprint | • | | | • | • | • | | | |
| | Facilitating behavior change | | | | • | | | | | • |
| | **Collective decisions** | | | | | | | | | |
| | Modeling social interactions | | | • | | • | | | | |
| | Informing policy | • | • | | • | | | | • | • |
| | Designing markets | | | | | • | • | | | • |
| | **Education** | | | | • | • | | | | |
| | **Finance** | | | | • | | • | | • | |

# Break up the empire

# Forge new paths


VOTE NO NOW
STOP PROJECT BLUE
NOT ONE DROP
PROJECT BLUE
AMAZON
PALANTIR
I.C.E.


¡RESISTIR!
Water for the people


SHARP
CIO AWARDS
CONFERENZ
FIGHTING FOR FAIR
servicenow
IDC
clearpoint
RWA
younity
juniper
CompTIA

When people rise…

empires fall.